

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

(192525277)

I G. Jaahnavi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Jaahnavi

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 & Type-2 hypervisors

Hypervisor :- It is a software that creates and manages multiple virtual machines on a single physical computer.

Type-1 Hypervisor :- It runs directly on the physical hardware without a host operating system.

Ex:- VMware ESXi, Microsoft Hyper-V

Type-2 Hypervisor :- It runs on top of a host operating system.

Ex:- Oracle VirtualBox

Feature	Type-1	Type-2
Runs on	Hardware	Host OS
performance	High	Moderate
Security	High	Low
Resource utilization	Excellent	Moderate
Scalability	High	Limited
Reliability	High	Moderate
Usage	Enterprise cloud data centers	Personal computers, testing

Performance :-

Type-1 hypervisors provide higher performance because they access hardware directly without a host operating system.

Type-2 hypervisors have additional overhead due to host OS, resulting in lower performance.

Security :- Type-1 hypervisors offer better security because there is no host operating system. Type-2 depends on the security of host operating system. If the host OS is compromised, the VMs can also be at risk.

Manageability :-

Type 1 hypervisors are designed for centralized data center management, resource allocation, monitoring, migration. Type 2 hypervisors are easier to install but suitable for development & desktop environments.

Justification :- For a mission critical cloud data center, Type-1 hypervisor is the better choice because it provides :

- High performance
- Better Security & isolation
- * efficient resource utilization.

Conclusion :-

Type-1 hypervisors are preferred for mission critical cloud environments because Performance, Security, reliability & manageability are more important than the simplicity offered by Type-2 hypervisors.

Virtualization architecture Supports isolation

2) Introduction :- Virtualization allows multiple operating systems to run independently on a shared physical server while maintaining complete isolation b/w workloads.

Workload isolation:-

1. VM isolation

- * Finance, healthcare, & student applications runs in separate VMs.

- * Failure in one VM does not affect others.

2. CPU Isolation

- * Hypervisor allocates dedicated CPU cores or virtual CPUs.

- * Prevents resource conflicts.

3. Memory Isolation

- * Each VM receives independent memory allocation.

- * Prevents unauthorized memory access.

4. Storage Isolation

- * virtual disks are separated.

- * Access permissions protect sensitive information

5. Network Isolation

- * VLANs & Software defined Networking (SDN) separate traffic.

- * Firewalls prevent unauthorized communication.

Major Attack Surfaces

① Hypervisor Attack

- * Attackers exploit vulnerabilities in hypervisor.
 - * may compromise all VMs hosted on the Server
- Protection
- * Frequent updates
 - * Secure Configuration
 - * Least privilege administration

② VM Escape Attack

- * Malware escapes from a guest VM on the host.
- * can access neighbouring VMs.

Protection

Patch management
Strong VM isolation
End point security.

Additional security Measures

- ① Multi-Factor Authentication (MFA)
- ② Encryption
- ③ security Information & Event Management (SIEM)

Conclusion :-

Virtualization architecture provides secure isolation through compute, memory & storage. proper security practices reduce risks from hypervisor & VM escape attacks.

3) VM Host utilization snapshot & recommend
Corrective actions.

CPU : 92 %

Memory : 88 %

Storage I/O Latency : 35 ms

Average VM Boot Time : 170 sec

Introduction:-

The given metrics indicate heavy resource utilization & performance degradation on the virtualization host.

Analysis

CPU utilization (92%)

- * CPU is nearly saturated
- * causes slow processing
- * Applications experience delays.

Memory utilization (88%)

- * Insufficient free RAM
- * Hypervisor may start swapping memory.
- * Reduces VM performance.

Storage Latency (35 ms)

- * Disk Subsystem is overloaded
- * slow read/write operations.
- * Database & applications performance decreases.

VM Boot Time (170 seconds)

- * Indicates CPU & Storage bottlenecks.
- * Longer recovery during failures.

Bottle necks :-

- * CPU overload.
- * Memory pressure
- * Slow storage

Corrective Actions

CPU

- * Add processors
- * Balance workloads
- * Live migration

Storage

- * Upgrade to SSD
- * Configure RAID
- * Use storage tiering.

Memory

- * Increase RAM
- * Enable memory
- * Remove unused VMs

General

- * Auto-scaling
- * Continuous monitoring
- * Load balancing.

Conclusion :-

Balanced allocation of CPU, memory, & Storage resources significantly improves cloud infrastructure performance.

4) SDDC architecture improves agility compared with a traditional datacenter.

Introduction :-

A Software Defined Datacenter virtualizes all infrastructure resources & manages them through software rather than manual hardware configuration.

Components

Compute virtualization

- * Multiple VMs per server
- * Dynamic resource allocation
- * Live migration.

Storage virtualization

- * Pools storage devices
- * Thin provisioning
- * Automatic backup.

Network virtualization

- * Creates logical networks
- * SDN enables Centralized Control.
- * Network Segmentation improves Security.

Advantages

- * Automation.
- * High availability
- * Elastic Scalability.

Comparison

Traditional Datacenter	Software Defined Datacenter
Hardware Centric infrastructure	Software controlled infrastructure
Manual resource allocation.	Automated resource allocation
slow deployment of application	Rapid application deployment
Limited Scalability	Highly Scalable & flexible
Higher maintaince cost	Lower operational cost
Manual Configuration	centralized Software management..

Applications:-

- * Cloud Service providers.
- * Enterprise IT
- * Banking
- * Healthcare

Conclusion :-

SDDC improves agility by virtualizing Compute, storage, & Networking, enabling Scalability, automation, & efficient resource utilization.

5) Multi cloud federation project fails due to identity mismatches b/w providers.

Introduction:-

Multi-cloud environments involve multiple cloud providers with different authentication mechanisms. Identity mismatches cause login failures & inconsistent access control.

Causes :-

- * Different identity providers.
- * Different authentication protocols.
- * Duplicate user accounts.
- * Inconsistent access policies.

Secure Identity Design

Identity Federation

- * Establish trust b/w cloud providers.
- * Share authentication information securely

Single sign-on (SSO)

- * users authenticate once
- * Access multiple cloud services seamlessly.

Identity Provider (IdP)

- * centralized user authentication.
- * Issues Secure Authentication tokens.

Federation Standards

- * SAML 2.0
- * OAuth 2.0
- * OpenID Connect

Privacy Protection

- * Encrypt identity information
- * Share only required user attributes
- * Implement Multi-Factor Authentication
- * Apply Role Based Access Control.

Benefits

- * Seamless user access
- * Strong authentication
- * Improved privacy.
- * Better user experience.

Conclusion :-

A secure federation architecture using SSO, Identity providers, SAML / OAuth / OpenID connect, MFA & RBAC resolves identity mismatches while ensuring secure & privacy preserving access across multiple cloud services providers.